5.Implementation of K-Means Clustering

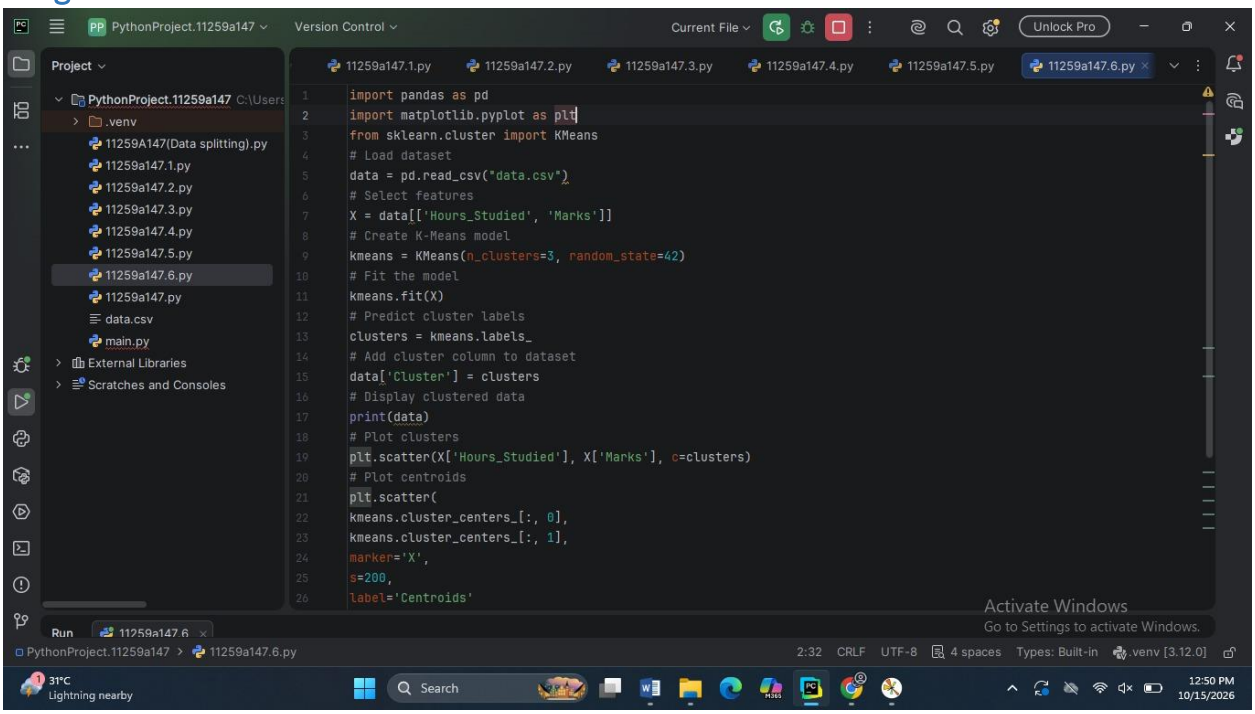
Aim

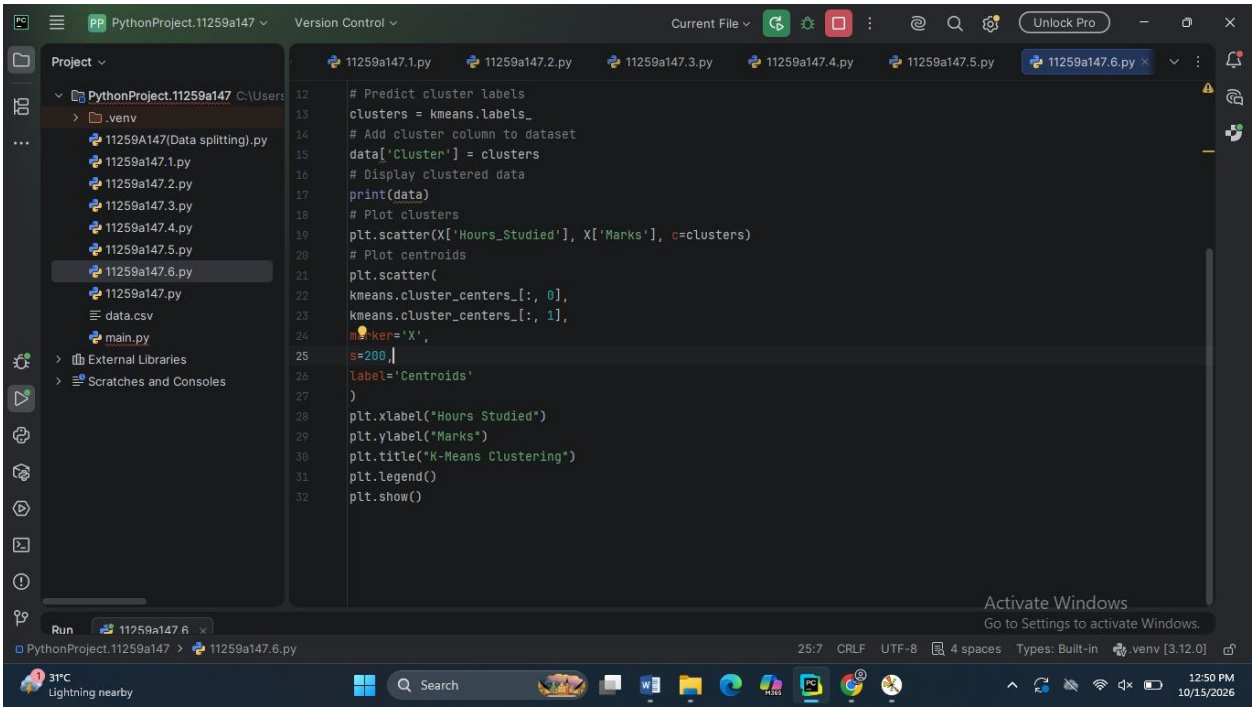
To write a program that implements the K-Means clustering algorithm to group the given data points into k clusters based on their similarity, and to display the cluster labels, the cluster centroids and a plot of the resulting clusters.

Algorithm:

1. Start the program.
2. Import the required libraries (pandas, numpy, matplotlib, and KMeans from sklearn.cluster).
3. Load the dataset from the input file.
4. Select the input features to be used for clustering (no target variable is needed, since clustering is unsupervised).
5. Choose the number of clusters k (for example k = 3).
6. Create the KMeans model with k clusters and a fixed random_state.
7. Randomly initialize k centroids.
8. Assign each data point to the nearest centroid using Euclidean distance.
9. Recalculate each centroid as the mean of all the data points assigned to that cluster.
10. Repeat steps 8 and 9 until the centroids no longer change or the maximum number of iterations is reached.
11. Fit the model on the data and obtain the cluster label for each data point.
12. Display the cluster labels and the final cluster centroids.
13. Plot the data points coloured by cluster, along with the centroids.
14. Stop the program.

Program:


PythonProject.11259a147
Version Control
Current File
Unlock Pro
Project
11259a147.1.py
11259a147.2.py
11259a147.3.py
11259a147.4.py
11259a147.5.py
11259a147.6.py
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
Load dataset
data = pd.read_csv("data.csv")
Select features
X = data[['Hours_Studied', 'Marks']]
Create K-Means model
kmeans = KMeans(n_clusters=3, random_state=42)
Fit the model
kmeans.fit(X)
Predict cluster labels
clusters = kmeans.labels_
Add cluster column to dataset
data['Cluster'] = clusters
Display clustered data
print(data)
Plot clusters
plt.scatter(X['Hours_Studied'], X['Marks'], c=clusters)
Plot centroids
plt.scatter(
kmeans.cluster_centers_[:, 0],
kmeans.cluster_centers_[:, 1],
marker='X',
s=200,
label='Centroids'
Activate Windows
Go to Settings to activate Windows.
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PythonProject.11259a147
Version Control
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Project
11259a147.1.py
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Predict cluster labels
clusters = kmeans.labels_
Add cluster column to dataset
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Display clustered data
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Plot clusters
plt.scatter(X['Hours_Studied'], X['Marks'], c=clusters)
Plot centroids
plt.scatter(
kmeans.cluster_centers_[:, 0],
kmeans.cluster_centers_[:, 1],
marker='X',
s=200,
label='Centroids'
)
plt.xlabel("Hours Studied")
plt.ylabel("Marks")
plt.title("K-Means Clustering")
plt.legend()
plt.show()
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Result:

```
Run  11259a147.6
C:\Users\Admin\PycharmProjects\PythonProject.11259a147\.venv\Scripts\python.exe C:\Users\Admin\PycharmProjects\PythonProject.11259a147\11259a147.6.py
  Student id   Hours_Studied  Attendance  Assignments  Marks Result  Cluster
0   11259a140              4          55           24     66   pass        2
1   11259a141              5          87           18     98   pass        0
2   11259a142              2          93           15     76   pass        2
3   11259a143              3          32           25     34   fail        1
4   11259a144              1          98           19     87   pass        0
```

Graph:

